

1. Define a class and an object in Java. Explain how to create classes and objects with a suitable example. Also discuss different types of access modifiers and their significance.
2. What is inheritance in Java? Explain its types with examples. Additionally, describe the concept of packages and their benefits in Java programming.
3. Discuss the concept of multithreading in Java. Explain the thread life cycle and demonstrate how to create a multithreaded program. Also explain object serialization with an example.
4. Develop a Java program where one thread produces data and another consumes it using some methods. Include proper exception handling mechanisms.
5. Create a multithreaded Java program to illustrate thread synchronization. Ensure that exceptions are handled gracefully.
6. Define an interface in Java. Create an interface `Employee` with methods `calculateSalary()` and `displayDetails()`. Implement it in classes `FullTimeEmployee` and `PartTimeEmployee`. Write a program to demonstrate the functionality and explain the execution.
7. Explain the concept of interfaces in Java. Design an interface `Animal` with methods `sound()` and `eat()`. Implement this interface in classes `Dog` and `Cat`. Write a complete program and describe how it works.
8. Write a Java program illustrating the scope and visibility of methods with different access modifiers.
9. Write a Java program to test accessibility of class members using different access modifiers. Include both variables and methods, and verify access rules across packages. Represent the results in a comparative table.
10. Define layout manager in Java. Explain any four types of layout managers and write example programs to demonstrate their usage.
11. Explain different layout managers used in Java AWT/Swing. Describe the working of any four layout managers with sample code snippets.
12. Develop a Java GUI program using JFC that accepts personal details (name, age, address) through input fields. Implement a `JavaBean` to store the data and display it in a formatted way when the user clicks a submit button.
13. Explain the architecture and working of the Java Event Delegation Model with a diagram.
14. Discuss different types of `JavaBean` properties (simple, indexed, bound, constrained).
15. Compare AWT and Swing (JFC) components in terms of features and performance.
16. Write a Java program to demonstrate event handling using `ActionListener`, `MouseListener`, or `KeyListener`.
17. List and explain different types of JDBC drivers used for RDBMS systems.
18. Explain the `java.sql` API and its key interfaces such as `Connection`, `Statement`, and `ResultSet`.
19. What is connection pooling? Explain its advantages and how it is implemented using the `javax.sql` package.
20. What is JDBC? Explain its role in connecting Java applications with relational databases.
21. Write a Java program to perform Insert, Select, Update, and Delete operations on a database table using JDBC.
22. Write a Java program to demonstrate database connectivity and perform basic operations like data insertion, retrieval, updation, and deletion.
23. Create a Java application that performs CRUD operations using `PreparedStatement` instead of `Statement`.
24. In a product inventory system, a user wants to insert product details (Product ID, Name, Price, Quantity) into a database. The program should accept multiple inputs and display all stored records. Implement this using JDBC and explain the logic.

25. Develop a Java Swing application using the MVC architecture to manage student details (add, view, update). Clearly separate Model, View, and Controller components.
26. Design a GUI-based calculator application using Swing and MVC pattern. Implement basic arithmetic operations and explain the role of each component (Model, View, Controller).
27. Develop a library management GUI application using Swing following the MVC design pattern. Include functionalities like adding and viewing books.
28. Write a Java program to implement a bank account management system using Swing and MVC architecture. The application should support deposit and withdrawal operations with proper separation of concerns.
29. Develop a Java Swing application to simulate an online job application form where users can select skills using checkboxes, choose job type using radio buttons, select location from a dropdown list, set experience level using a slider, and display the submitted information.
30. Create a Java Swing GUI for a survey form where users can select interests using checkboxes, choose age group using radio buttons, select country from a combo box, rate satisfaction using a slider, and view results upon submission.
31. Design a Java Swing application for a feedback system where users can select services using checkboxes, choose feedback type using radio buttons, select branch location from a dropdown, rate services using a slider, and display the feedback summary.
32. Develop a Swing-based notepad application featuring menus (File, Edit), a toolbar for quick actions, and dialog boxes for opening, saving, and exiting the application.
33. Provide a detailed comparison between AWT and Swing. Also, write short notes on five components each (such as Button, Label, TextField, Frame for AWT and JButton, JLabel, JTextField, JFrame, JTable for Swing) with examples.
34. Write a Java program to implement a client-server communication system where the client sends a message and the server responds with an acknowledgment.
35. Create a Java program to implement a two-way communication between client and server using sockets.
36. Develop a Java socket program where the client sends a number to the server and the server returns its square or factorial.
37. Develop a Java socket program where multiple clients can connect to a server and exchange messages.
38. Write a Java program to demonstrate TCP-based client-server communication using sockets and explain its working.
39. Write a Java Applet to draw different geometric shapes such as square, ellipse, line, and polygon, and display a message "Geometric Figures". Use appropriate coordinates. Also write the HTML code to execute the applet.
40. Develop a Java Applet that draws circle, rectangle, triangle, and arc with different colors and displays a string "Drawing Shapes". Provide suitable coordinates and include the HTML code to run the applet.
41. Create a Java Applet program to draw multiple shapes like line, oval, rounded rectangle, and arc, and display the text "Graphics Demo". Also write the corresponding HTML code to embed and run the applet.
42. What is the role of an EJB container?
43. What is EJB (Enterprise JavaBeans)? Mention its purpose in enterprise applications.
44. Write a Java EJB program to implement a stateful session bean that maintains user session data (e.g., shopping cart items).
45. Develop an EJB application for a banking system using a stateful session bean to handle deposit and withdrawal operations.

46. Write a Java program to demonstrate single inheritance where a class Student inherits properties from a class Person. Display details of the student.
47. Develop a Java program to demonstrate multilevel inheritance using classes Animal → Mammal → Dog. Implement methods to show characteristics at each level and explain the program execution.
48. Develop a Java program to handle multiple exceptions using multiple catch blocks.
49. Discuss the differences between Socket programming, Remote Method Invocation (RMI), and Servlet-based communication. Illustrate suitable scenarios where each technique is preferred.
50. Compare Sockets, RMI, and Servlets in terms of architecture, performance, scalability, and ease of development. Explain their use in real-world applications.